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Course: Python Programming
Week: 3

1) Bank Account Class

- **Create a class with methods: deposit, withdraw, display balance.**

Program:

```
class BankAccount:
    def __init__(self, balance=0):
        self.balance = balance
    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)
    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
        else:
            print("Insufficient balance")
    def display_balance(self):
        print("Balance:", self.balance)

balance = float(input("Enter initial balance: "))
account = BankAccount(balance)
deposit = float(input("Enter deposit amount: "))
account.deposit(deposit)
withdraw = float(input("Enter withdrawal amount: "))
account.withdraw(withdraw)
account.display_balance()
```

Output:

```
Enter initial balance: 500
Enter deposit amount: 200
Deposited: 200.0
Enter withdrawal amount: 150
Withdrawn: 150.0
Balance: 550.0
```

2)Library Management System (OOP)

- Add/remove books, issue/return books.

```
class Library:
    def __init__(self):
        self.books = []
    def add_book(self, book):
        self.books.append(book)
        print("Book added:", book)
    def remove_book(self, book):
        if book in self.books:
            self.books.remove(book)
            print("Book removed:", book)
        else:
            print("Book not found")
    def issue_book(self, book):
        if book in self.books:
            self.books.remove(book)
            print("Book issued:", book)
        else:
            print("Book not available")
    def return_book(self, book):
        self.books.append(book)
        print("Book returned:", book)
    def display_books(self):
        print("Available books:", self.books)

library = Library()
book = input("Enter book to add: ")
library.add_book(book)
book = input("Enter another book to add: ")
library.add_book(book)
library.display_books()
book = input("Enter book to issue: ")
library.issue_book(book)
library.display_books()
book = input("Enter book to return: ")
library.return_book(book)
library.display_books()
book = input("Enter book to remove: ")
library.remove_book(book)
library.display_books()
```

Output:

```
Enter book to add: py
Book added: py
Enter another book to add: ja
Book added: ja
Available books: ['py', 'ja']
Enter book to issue: py
Book issued: py
Available books: ['ja']
Enter book to return: ja
Book returned: ja
Available books: ['ja', 'ja']
Enter book to remove: ja
Book removed: ja
Available books: ['ja']
```

3) Calculator Class with Exception Handling**Program:**

```
class Calculator:
    def add(self, a, b):
        return a + b
    def subtract(self, a, b):
        return a - b
    def multiply(self, a, b):
        return a * b
    def divide(self, a, b):
        try:
            return a / b
        except ZeroDivisionError:
            return "Cannot divide by zero"
calc = Calculator()
try:
    a = float(input("Enter first number: "))
    b = float(input("Enter second number: "))
    print("Addition:", calc.add(a, b))
    print("Subtraction:", calc.subtract(a, b))
    print("Multiplication:", calc.multiply(a, b))
    print("Division:", calc.divide(a, b))
except ValueError:
```

```
print("Please enter numbers only")
```

Output:

Enter first number: 24

Enter second number: 25

Addition: 49.0

Subtraction: -1.0

Multiplication: 600.0

Division: 0.96